CSE 242 Assignment 1 — Ioannis Barous

Machine Learning, UC Santa Cruz, Fall 2026 — Derivations

October 16, 2026

Final answers are **boxed**. The numbers marked *sample* in Part 3 come from running `Assignment1.py` with its default settings ($n = 500$ samples, seed 0).

Part 1: Linear Algebra

Q1. Matrix operations

$$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad (XY)_{ij} = \sum_k X_{ik} Y_{kj}.$$

AB**:**

$$AB = \begin{bmatrix} 3\cdot 1 + 4\cdot 0 & 3\cdot 2 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 2 + 1\cdot(-1) \end{bmatrix} = \boxed{\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}}$$

BA**:**

$$BA = \begin{bmatrix} 1\cdot 3 + 2\cdot 2 & 1\cdot 4 + 2\cdot 1 \\ 0\cdot 3 + (-1)\cdot 2 & 0\cdot 4 + (-1)\cdot 1 \end{bmatrix} = \boxed{\begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}}$$

Since $AB \neq BA$, matrix multiplication is not commutative.

$(A + B)^T$**:**

$$A + B = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix} \quad \Longrightarrow \quad (A+B)^T = \boxed{\begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}}$$

AB^T**:** with $B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$,

$$AB^T = \begin{bmatrix} 3\cdot 1 + 4\cdot 2 & 3\cdot 0 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 2 & 2\cdot 0 + 1\cdot(-1) \end{bmatrix} = \boxed{\begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}}$$

Traces (sum of the diagonal entries):

$$\operatorname{trace}(A) = 3 + 1 = \boxed{4}, \qquad \operatorname{trace}(AB) = 3 + 3 = \boxed{6}.$$

Check: $\text{trace}(BA) = 7 + (-1) = 6 = \text{trace}(AB)$, as Q3 proves in general.

Q2. Eigenvalues and eigenvectors

$$M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Characteristic polynomial:

$$\det(M - \lambda I) = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0.$$

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 3}$$

Check: $\lambda_1 + \lambda_2 = 4 = \text{trace}(M)$ and $\lambda_1 \lambda_2 = 3 = \det(M)$.

Eigenvectors, solving $(M - \lambda I)v = 0$:

- $\lambda_1 = 1$: $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} v = 0 \Rightarrow v_1 = v_2$, so $\boxed{v^{(1)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$.
- $\lambda_2 = 3$: $\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} v = 0 \Rightarrow v_2 = -v_1$, so $\boxed{v^{(2)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$.

The two eigenvectors are orthogonal, $[1, 1] \cdot [1, -1] = 0$, as expected for a symmetric matrix.

Verification of $Mv = \lambda v$:

$$M \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad M \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2+1 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Q3. Proof that $\text{trace}(AB) = \text{trace}(BA)$

Let A be $m \times n$ and B be $n \times m$. Then AB is $m \times m$ and BA is $n \times n$, so both traces are defined.

1. The diagonal entries are $(AB)_{ii} = \sum_{k=1}^{n} A_{ik} B_{ki}$ and $(BA)_{kk} = \sum_{i=1}^{m} B_{ki} A_{ik}$.
2. By definition of the trace,

$$\text{trace}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{k=1}^{n} A_{ik} B_{ki}.$$

3. The sum is finite, so the order of summation can be swapped. Scalars commute, so $A_{ik} B_{ki} = B_{ki} A_{ik}$:

$$\text{trace}(AB) = \sum_{k=1}^{n} \sum_{i=1}^{m} B_{ki} A_{ik}.$$

4. The inner sum is $(BA)_{kk}$, so

$$\boxed{\operatorname{trace}(AB) = \sum_{k=1}^{n} (BA)_{kk} = \operatorname{trace}(BA)} \qquad \blacksquare$$

This holds even though AB and BA may have different sizes. Example from Q1: both traces equal 6.

Q4. Inverse of C

$$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

Existence. A square matrix is invertible if and only if its determinant is nonzero:

$$\det(C) = 4 \cdot 6 - 7 \cdot 2 = 24 - 14 = 10 \neq 0 \quad \Longrightarrow \quad C \text{ is invertible.}$$

Adjugate formula. For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc \neq 0$, the inverse is $\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. This swaps the diagonal entries and negates the off-diagonal ones:

$$C^{-1} = \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \boxed{\begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}}$$

Verification:

$$C\,C^{-1} = \begin{bmatrix} 4(0.6) + 7(-0.2) & 4(-0.7) + 7(0.4) \\ 2(0.6) + 6(-0.2) & 2(-0.7) + 6(0.4) \end{bmatrix} = \begin{bmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Part 2: Probability, MLE, and MAP Estimation

Q1. Normalization, expectation, and variance

$P(X = k) = c\,\dfrac{r^k}{k!}$ for $k = 0, 1, 2, \ldots$, with $r = 3$.

(a) Normalization. The probabilities must sum to 1. Using the exponential series $\sum_{k \geq 0} r^k/k! = e^r$:

$$\sum_{k=0}^{\infty} c\,\frac{r^k}{k!} = c\,e^r = 1 \quad \Longrightarrow \quad c = e^{-r} = \boxed{e^{-3} \approx 0.0498}$$

So $X \sim \text{Poisson}(r)$.

(b) Expectation. The $k = 0$ term vanishes, and $k/k! = 1/(k-1)!$. Substitute $j = k - 1$:

$$E[X] = \sum_{k=1}^{\infty} k\, e^{-r}\frac{r^k}{k!} = e^{-r}\, r \sum_{j=0}^{\infty} \frac{r^j}{j!} = e^{-r}\, r\, e^{r} = r = \boxed{3}$$

(b) Variance, using the $E[X(X-1)]$ trick. The $k = 0, 1$ terms vanish, and $k(k-1)/k! = 1/(k-2)!$. Substitute $j = k - 2$:

$$E[X(X-1)] = e^{-r} \sum_{k=2}^{\infty} \frac{r^k}{(k-2)!} = e^{-r}\, r^2 \sum_{j=0}^{\infty} \frac{r^j}{j!} = r^2.$$

$$E[X^2] = E[X(X-1)] + E[X] = r^2 + r, \qquad \mathrm{Var}(X) = E[X^2] - (E[X])^2 = r^2 + r - r^2 = r = \boxed{3}$$

The last step uses the identity proved in Q6.

Q2. Event probabilities

Given $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \mid B) = 0.6$.

(a) Product rule:

$$P(A \cap B) = P(A \mid B)\, P(B) = 0.6 \times 0.3 = \boxed{0.18}$$

(b) Bayes' rule:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.5} = \boxed{0.36}$$

(c) Inclusion–exclusion. $A \cap B$ is counted twice in $P(A) + P(B)$, so subtract it once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.18 = \boxed{0.62}$$

Sanity checks: $0.18 \le \min(0.5, 0.3)$ and $0.5 \le 0.62 \le 1$.

Q3. Coin: MLE and MAP

Observed H, H, T, H, T, so $n = 5$ tosses with $h = 3$ heads and $n - h = 2$ tails. The tosses are i.i.d. $\mathrm{Bernoulli}(\theta)$.

(a) MLE. The likelihood and log-likelihood are

$$L(\theta) = \theta^h (1-\theta)^{n-h} = \theta^3 (1-\theta)^2, \qquad \ell(\theta) = h \log\theta + (n-h)\log(1-\theta).$$

Set the derivative to zero for $0 < \theta < 1$:

$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \implies h(1-\theta) = (n-h)\theta \implies \hat{\theta}_{\mathrm{MLE}} = \frac{h}{n}.$$

The second derivative $-h/\theta^2 - (n-h)/(1-\theta)^2 < 0$, so this is a maximum.

$$\boxed{\hat{\theta}_{\text{MLE}} = \tfrac{3}{5} = 0.6}$$

Boundary cases. The log is undefined at 0 and 1, so check L directly:

- If $h = 0$, $L = (1-\theta)^n$ is decreasing, so the maximum is at $\theta = 0$.
- If $h = n$, $L = \theta^n$ is increasing, so the maximum is at $\theta = 1$.

Both agree with h/n.

(b) MAP with prior $p(\theta) = 2\theta$ on $[0, 1]$. This is a valid density because $\int_0^1 2\theta \, d\theta = 1$. The posterior is

$$p(\theta \mid \text{data}) \propto L(\theta)\, p(\theta) \propto \theta^{h+1}(1-\theta)^{n-h},$$

which is a $\text{Beta}(h+2,\ n-h+1)$ shape. Set the derivative of the log-posterior to zero:

$$\frac{h+1}{\theta} - \frac{n-h}{1-\theta} = 0 \implies (h+1)(1-\theta) = (n-h)\theta \implies \hat{\theta}_{\text{MAP}} = \frac{h+1}{n+1}.$$

The second derivative is negative, so this is a maximum.

$$\boxed{\hat{\theta}_{\text{MAP}} = \tfrac{4}{6} = \tfrac{2}{3} \approx 0.667}$$

The prior favours larger θ, so the MAP lies above the MLE.

Boundary cases:

- If $h = n$, the posterior $\propto \theta^{n+1}$ is increasing, so the maximum is at $\theta = 1 = (n+1)/(n+1)$.
- If $h = 0$, the posterior $\propto \theta(1-\theta)^n$ is zero at both ends, so the maximum is the interior point $1/(n+1)$. It is not at 0, because the prior vanishes there.

So $\hat{\theta}_{\text{MAP}} = (h+1)/(n+1)$ for every $0 \le h \le n$.

Q4. Gaussian MLE for $X = \{1, 3, 5, 7\}$

With $x_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Mean:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_i (x_i - \mu) = 0 \implies \hat{\mu} = \frac{1}{n}\sum_i x_i.$$

Variance. Treat $s = \sigma^2$ as the variable:

$$\frac{\partial \ell}{\partial s} = -\frac{n}{2s} + \frac{1}{2s^2}\sum_i (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \hat{\mu})^2.$$

The MLE uses divisor n, not $n-1$. The $n-1$ version is the unbiased estimator; the MLE is slightly biased low.

Plug in $n = 4$:

$$\hat{\mu} = \frac{1+3+5+7}{4} = \boxed{4}, \qquad \hat{\sigma}^2 = \frac{(-3)^2 + (-1)^2 + 1^2 + 3^2}{4} = \frac{20}{4} = \boxed{5}.$$

Q5. Gaussian MAP for the mean

Likelihood $x \mid \mu \sim \mathcal{N}(\mu, s^2)$ with known $s^2 = 4$, and prior $\mu \sim \mathcal{N}(m_0, t^2)$ with $m_0 = 0$ and $t^2 = 1$. One observation, $x = 5$.

Posterior $\propto$ likelihood $\times$ prior:

$$\log p(\mu \mid x) = -\frac{(x-\mu)^2}{2s^2} - \frac{(\mu - m_0)^2}{2t^2} + \text{const}.$$

Set the derivative to zero:

$$\frac{x-\mu}{s^2} - \frac{\mu - m_0}{t^2} = 0 \implies \hat{\mu}_{\text{MAP}} = \frac{x/s^2 + m_0/t^2}{1/s^2 + 1/t^2}.$$

The second derivative $-(1/s^2 + 1/t^2) < 0$, so this is a maximum. Equivalently, completing the square shows the posterior is Gaussian with precision $1/s^2 + 1/t^2$, and its mode equals its mean. The result is a precision-weighted average of the observation and the prior mean.

Plug in:

$$\hat{\mu}_{\text{MAP}} = \frac{5/4 + 0/1}{1/4 + 1/1} = \frac{1.25}{1.25} = \boxed{1}$$

The prior has precision 1 and the observation has precision $1/4$, so the prior is four times more confident. That is why the estimate is pulled strongly from 5 toward 0.

Q6. Proof that $\text{Var}(X) = E[X^2] - (E[X])^2$

Let $\mu = E[X]$, a constant. Then

$$
\begin{aligned}
\mathrm{Var}(X) &= E\big[(X-\mu)^2\big] && \text{(definition)}\\
&= E\big[X^2 - 2\mu X + \mu^2\big] && \text{(expand the square)}\\
&= E[X^2] - 2\mu\,E[X] + \mu^2 && \text{(linearity of expectation)}\\
&= E[X^2] - 2\mu^2 + \mu^2 && (E[X]=\mu)\\
&= E[X^2] - \mu^2.
\end{aligned}
$$

$$\boxed{\mathrm{Var}(X) = E[X^2] - (E[X])^2} \qquad \blacksquare$$

Part 3: Principal Component Analysis

Q1. Variance of a projection

Let $X \in \mathbb{R}^d$ be a random vector and u a fixed unit vector, $\|u\| = 1$.

Definitions:

$$\mu = E[X], \qquad \Sigma = E\big[(X-\mu)(X-\mu)^T\big], \qquad \Sigma_{ij} = \mathrm{Cov}(X_i, X_j).$$

Σ is symmetric and positive semi-definite.

Derivation:

1. By linearity, $E[u^T X] = u^T E[X] = u^T \mu$.
2. By the definition of variance, $\mathrm{Var}(u^T X) = E\big[(u^T X - u^T \mu)^2\big] = E\big[(u^T(X-\mu))^2\big]$.
3. $u^T(X-\mu)$ is a scalar, so it equals its transpose $(X-\mu)^T u$. Hence $(u^T(X-\mu))^2 = u^T(X-\mu)(X-\mu)^T u$.
4. u is constant, so it can be moved outside the expectation: $\mathrm{Var}(u^T X) = u^T\, E\big[(X-\mu)(X-\mu)^T\big]\, u$.
5. The middle term is Σ:

$$\boxed{\mathrm{Var}(u^T X) = u^T \Sigma\, u}$$

Since a variance is non-negative, $u^T \Sigma u \geq 0$ for all u. So Σ is PSD and its eigenvalues are ≥ 0.

Why the top eigenvector maximizes the projected variance. We want to maximize $u^T \Sigma u$ subject to $u^T u = 1$. The Lagrangian is $J(u, \lambda) = u^T \Sigma u - \lambda(u^T u - 1)$, and

$$\nabla_u J = 2\Sigma u - 2\lambda u = 0 \implies \Sigma u = \lambda u, \qquad u^T \Sigma u = \lambda\, u^T u = \lambda.$$

Every stationary point is an eigenvector, and the variance along it equals its eigenvalue. So the maximum is reached at the eigenvector with the **largest eigenvalue**: the first principal component. Adding the constraint "orthogonal to the earlier components" gives the next eigenvectors in decreasing order of eigenvalue.

Q2. Synthetic correlated Gaussian data

The data are drawn as $X \sim \mathcal{N}(m, S)$ with

$$m = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad S = \begin{bmatrix} 3 & 1.2 \\ 1.2 & 1 \end{bmatrix}.$$

S is positive definite, since $\det S = 3 - 1.44 = 1.56 > 0$ and $3 > 0$. NumPy's `default_rng(seed).multivariate_normal` factorizes $S = LL^T$ and returns $m + Lz$ with $z \sim \mathcal{N}(0, I)$. The same seed always gives the same samples.

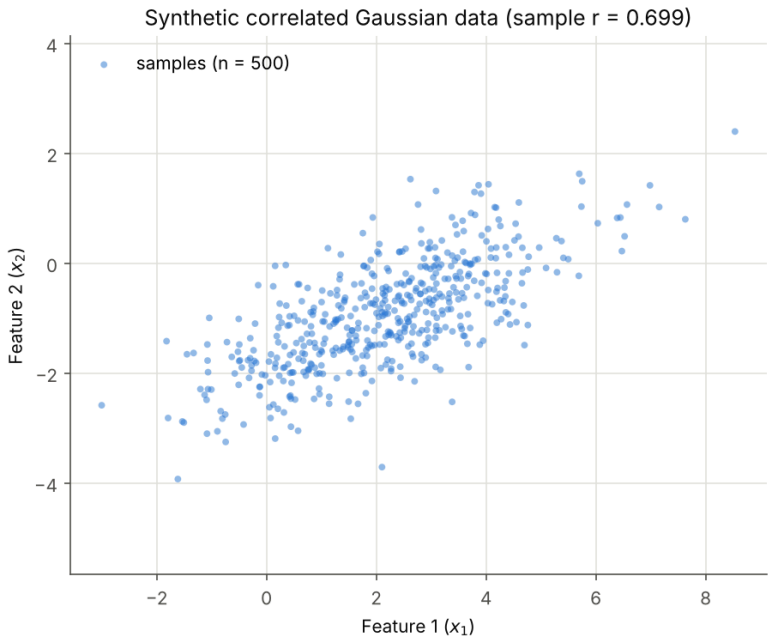
Correlation:

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}} = \frac{1.2}{\sqrt{3 \cdot 1}} \approx \boxed{0.693 \text{ (theory)}}, \qquad \hat{\rho} \approx \boxed{0.699 \text{ (sample)}}.$$

Sample estimates versus the true values:

Quantity	Sample ($n = 500$)	True
Mean	$[2.152,\ -0.935]$	$[2,\ -1]$
$\mathrm{Var}(X_1)$, $\mathrm{Var}(X_2)$	2.916, 0.953	3, 1
$\mathrm{Cov}(X_1, X_2)$	1.166	1.2
Correlation	0.699	0.693

The scatter plot is an ellipse tilted upward to the right: when one feature is above its mean, the other tends to be too. The ellipse is wider along x_1 (variance 3) than along x_2 (variance 1), so its long axis is tilted only about 25°, not 45°. The small differences from the true values are sampling noise, which shrinks like $1/\sqrt{n}$.



Synthetic correlated Gaussian data (n = 500, seed = 0).

Q3. PCA from scratch, linked to Q1

Step	Computation	Link to Q1
1. Center	$X_c = X - \bar{x}$	estimates $X - \mu$
2. Covariance	$\hat{\Sigma} = \frac{1}{n-1} X_c^T X_c$	sample version of $E[(X-\mu)(X-\mu)^T]$; $n-1$ makes it unbiased
3. Eigendecomposition	$\hat{\Sigma} u = \lambda u$ via `np.linalg.eigh`	the Lagrange condition; $\hat{\Sigma}$ is symmetric, so the eigenvectors are orthonormal
4. Select	sort λ in descending order	$u^T \hat{\Sigma} u = \lambda$ is the variance along u
5. Project	scores $= X_c U$	coordinates $u_k^T x_c$; score k has variance λ_k

Sample results:

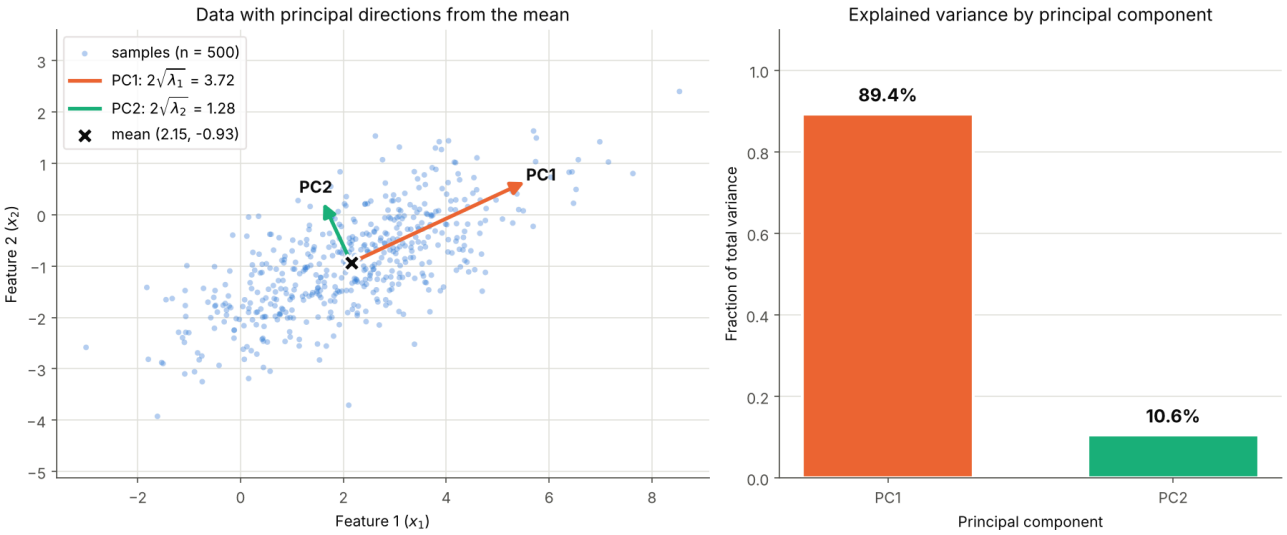
$$\hat{\lambda} \approx \boxed{[3.458,\ 0.411]}\ (\text{true } [3.562,\ 0.438]), \qquad \text{PC1} \approx \begin{bmatrix} 0.907 \\ 0.422 \end{bmatrix},\ \text{PC2} \approx \begin{bmatrix} -0.422 \\ 0.907 \end{bmatrix}.$$

The true eigenvalues come from $\lambda = 2 \pm \sqrt{2.44}$. Signs of the eigenvectors are arbitrary. Rank-deficient data, such as points on a line, give $\lambda_2 = 0$; any tiny negative value from round-off is clipped to 0 because Σ is PSD. Tied eigenvalues, as in an isotropic cloud, give a valid orthonormal basis for the tied eigenspace.

The plot. The left panel shows the samples, the mean (at about $(2.15, -0.93)$) and arrows of length $2\sqrt{\lambda_k}$, which is two standard deviations along each component:

- PC1 has length 3.72 and follows the long axis of the ellipse, about 25° above x_1.
- PC2 has length 1.28 and follows the short axis, perpendicular to PC1.

The axes use equal aspect, so the right angle is drawn correctly. This is exactly Q1: the direction maximizing $u^T \Sigma u$ is the top eigenvector, and the spread along each arrow is $\sqrt{\lambda}$.



Left: data with principal directions drawn from the mean. Right: explained variance per component.

Q4. Explained variance and dimensionality reduction

The total variance is $\operatorname{trace}(\Sigma) = \sum_j \lambda_j$, because the trace is unchanged by the orthogonal change of basis. Component k therefore explains

$$\text{fraction}_k = \frac{\lambda_k}{\sum_j \lambda_j}.$$

$$\boxed{\text{PC1} \approx 89.4\%, \qquad \text{PC2} \approx 10.6\%} \qquad (\text{theory } 89.1\% \text{ and } 10.9\%).$$

Why it matters. Keeping only PC1 (projecting $2\text{D} \to 1\text{D}$) keeps about 89% of the variance and stores one number per sample instead of two. The discarded variance $\lambda_2 \approx 0.411$ equals the squared reconstruction error summed over samples, divided by $n - 1$. By Q1, no other 1-D linear projection keeps more variance, so none has a smaller reconstruction error.

Tradeoff.

- Fewer components give more compression, remove noise and make downstream models faster, but lose more information.
- More components give higher fidelity but less reduction.

- A common rule is to keep the smallest k whose cumulative explained variance reaches a threshold such as 90–95%. Here one component gives about 89%, just under 90%. Two components give 100% but no reduction.
- The benefit is modest in 2-D but large in high dimensions, where a few components often capture most of the variance.
- PCA only captures linear structure and is sensitive to feature scale. Standardize the features first when their units differ.